

[Thoshith R R 250701797-2025-CSE](#) ▾**T2****Started on** Friday, 21 August 2026, 4:16 PM**State** Finished**Completed on** Friday, 21 August 2026, 4:16 PM**Time taken** 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int arr[], int low, int high)
4  {
5      if (low > high)
6          return 0;
7
8      if (arr[high] == 1)
9          return 0;
10
11     if (arr[low] == 0)
12         return high - low + 1;
13
14     int mid = (low + high) / 2;
15
16     if (arr[mid] == 0)
17         return high - mid + 1;
18
19     return countZeroes(arr, mid + 1, high);
20 }
21
22 int main()
23 {
24     int m;
25     scanf("%d", &m);
26
27     int arr[m];
28
29     for (int i = 0; i < m; i++)
30         scanf("%d", &arr[i]);
31
32     printf("%d", countZeroes(arr, 0, m - 1));
33
34     return 0;
35 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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